A Novel Multidimensional Domain Deep Learning Network for SAR Ship Detection

Dong Li, *Senior Member, IEEE*, Quanhuan Liang, Hongqing Liu, *Senior Member, IEEE*, Qinghua Liu, Haijun Liu, and Guisheng Liao, *Senior Member, IEEE*

***Abstract*— Since only the spatial feature information of ship target is utilized, the current deep learning-based synthetic aperture radar (SAR) ship detection approaches cannot achieve a satisfactory performance, especially in the case of multiscale or rotations, and the complex background. To overcome these issues, a novel multidimensional domain deep learning network for SAR ship detection is developed in this work to exploit the spatial and frequency-domain complementary features. The proposed method consists of the following main three steps. First, to learn hierarchical spatial features, the feature pyramid network (FPN) is adopted to produce ship target spatial multiscale characteristics with a top-down structure. Second, with a polar Fourier transform, the rotation-invariant features of SAR ship targets are obtained in the frequency domain. After that, a novel spatial-frequency characteristics fusion network is then presented, which seeks to learn more compact feature representations across different domains by updating the parameters of sub-networks interactively. The detection results are obtained due to utilizing the multidimensional domain information, and we evaluate the effectiveness of the proposed method using the existing SAR ship detection data set (SSDD). The results of the proposed method outperform other convolutional neural network (CNN)-based algorithms, especially for multiscale and rotation ship targets under complex backgrounds.**

***Index Terms*— Deep learning, multidimensional domain features, ship detection, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) has been extensively applied in many civilian and military fields due to its ability to produce high-resolution microwave images of observed scenes in all-day and all-weather conditions [1]–[3]. In recent years, SAR ship detection also receives increasing the attentions for its vital values in practical applications such as maritime management, marine traffic control, and marine environmental protection [4]–[6]. However, the existing SAR ship detection methods cannot produce a satisfactory performance because only the shallow features of targets are explored, and moreover, these methods mainly rely on manual participations. With the rapid developments of SAR imaging technology, the detailed structures of ship targets can be precisely produced, and therefore, it is possible to perform SAR ship detection based on refined information and high-level features.

Over the years, many methods have been proposed to detect SAR ship target, and these detection methods are mainly classified into three groups via exploiting the backscatter-characteristics, polarization-characteristics, and geometric-characteristics [7]–[9]. In the first group, the constant false alarm rate (CFAR) detection algorithms [10] and its extended versions [11]–[13] have been widely adopted in SAR ship detection. The CFAR method adaptively adjusts the detection threshold according to the established statistical distribution model of background clutter, and the most appropriate threshold can be obtained for the ship target. However, it is well-known that developing an accurate statistical model is a challenging task, and these models are vulnerable to wave and environmental changes. Moreover, they also present great difficulties for detecting ship targets near ports and reefs. In the second group, the polarization-characteristics-based methods make a use of the differences in scattering mechanisms between ship target and sea clutter to distinguish target and background [14], [15]. In recent years, with the launch of PolSAR satellite, the detection performance of PolSAR has been significantly improved due to the abundant scattering information compared with that of single-polarization SAR images. However, it is an enormous task to establish a comprehensive library of polarization scattering characteristics. In addition, the polarization characteristics-based methods have a high requirement for detecting environments and they can be easily affected by sea clutter background. The last group is geometric-characteristics-based methods by using the shape, area, size, and texture features [16], [17], where the

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template matching strategy is usually adopted to conduct the ship detections. Since these features directly reflect the physical characteristics of SAR ship target, and they have become the dominant SAR ship target detection methods. However, the target detection performance of those methods largely depends on the template library established using expert experiences. Moreover, they need to match all SAR image pixels with the template, which brings a huge computation cost, and due to the poor transfer ability, they are difficult to be applied to other ship detection scenarios. To summarize, traditional SAR ship detection methods only extract simple low-level features and they rely on manual participations, which requires selecting target features and determining the feature vector dimensions. Therefore, it is obvious that an accurate and robust ship target detection method in high-resolution SAR images needs to be further developed.

In recent studies, with the rise of artificial intelligence and the rapid developments of SAR imaging technology, it is possible to further improve SAR ship target detection accuracy by exploiting the refined information and high-level features of targets. Currently, several deep learning-based SAR ships detection methods have been proposed, and these methods can be divided into two categories of region convolutional neural network (RCNN)-based two-stage detector [20] and you only look once (YOLO)-based one-stage detector [27]–[29]. By taking the advantage of the low-level texture and high-level semantic features, a modified Faster RCNN method called as multilayer fusion light-head detector (MFLHD) is developed to improve the ship target detection performances [18]. In [19], a squeeze and excitation rank Faster RCNN (SER Faster-RCNN) method is proposed, and compared with the existing RCNN method, it obtains a better detection performance by learning the target's multiscale characteristics and introducing the squeeze and excitation mechanism. In [21] and [22], the attention mechanisms are also applied into the feature pyramid network (FPN) [23] to obtain more refined multiscale feature expressions. For the YOLO-based one-stage detector, in [24], a simplified U-Net [25] is proposed to achieve a low-cost SAR ship detection, but the detection performance is unsatisfactory. In [26], the DarkNet-53 is adopted to combine multiscale features into a single-shot multibox detector (SSD) network to improve multiscale ship detection performance. Compared with the RCNN-based two-stage detector, although a one-stage detector promotes the detection speed, it is at the expense of detection accuracy. Moreover, in practical applications, the SAR ship targets present the complex rotation motions and they are also vulnerable to sea clutter and inshore background. Therefore, the current deep learning-based SAR ship detection methods cannot achieve a satisfactory performance since only the spatial feature information of ship target is utilized.

To address the abovementioned issues, in this work, we propose a novel multidimensional domain deep learning network for SAR ship detection, which exploits the spatial and frequency-domain complementary features. First, to improve the performance of detecting multiscale ships in complex SAR scenes, FPN with a top-down structure using lateral connections is utilized to produce the multiscale characteristics. Second, the frequency-domain features that are not sensitive to ship target rotation are obtained by the polar Fourier transform [30]. Finally, we present a novel spatial-frequency features fusion network that achieves the cross-domain fusion of features by updating the parameters of different sub-networks interactively to enhance the multidimensional domain feature representations. The effectiveness and feasibility of the proposed method using SAR ship detection data set (SSDD) [31] are verified and the experiments show that the proposed approach outperforms other Convolutional Neural Network (CNN)-based algorithms, especially for multiscale and rotation ship targets under complex backgrounds.

The remaining part of this article proceeds as follows. In Section II, the detailed developments of the proposed method are introduced. The experimental results and analysis of the method using SSDD are shown in Section III. Finally, Section IV is a summary of our work.

II. Proposed Method Descriptions

In this section, the multidimensional domain deep learning network is developed, and details of the implementation procedures are also presented. To that aim, the motivations and the overall structure of the network are illustrated by analyzing the shortcomings of existing methods. After that, the implementations of the spatial feature extraction, frequency-domain feature extraction, and feature fusion are described in detail. At the end of this section, the loss function of the proposed method is described.

A. Problem Formation and Method Overview

Addressing the problem of multiscale and rotation of objects plays an important role in ship detection. Especially, the SAR image tends to have a complex rotational behavior and scale variation because of the "bird perspective," shown in Fig. 1. Fig. 1(a) shows several examples of multiscale ships in SAR images and Fig. 1(b) depicts examples of ship rotations in SAR images. The current CNN-based methods mainly focus on the scale transformation of ship target, but the rotation behavior of the target is sensitive in the spatial domain. They usually solve the rotation problem by rotating SAR images or convolution kernels at different angles. However, it cannot completely overcome this issue due to the influence of the pre-setting rotation angles. Additionally, only the single-dimensional domain spatial feature information is utilized, which results in the limited performances of SAR ship detection.

In this work, we propose a novel deep learning network for SAR images by employing the complementary information of multidimensional domain for SAR ship detection to effectively address the aforementioned issues. The processing scheme of our method is shown in Fig. 2. This is a two-stage object detection network. First, the backbone network and FPN are adopted to extract multiscale features of ship target. Next, the rotation-invariant spectrum features are produced through the polar Fourier transform. The feature fusion network is then developed to refine the spatial-frequency features and realize the cross-domain fusion of multidimensional domain features. The proposed network comprehensively utilizes the

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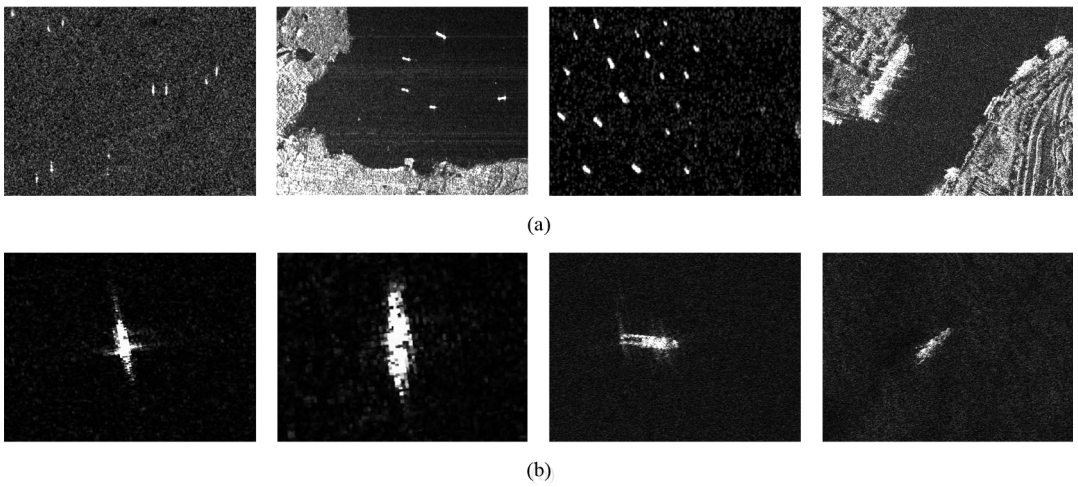

Fig. 1. Several SAR images from the SSDD data set. (a) Examples of multiscale ships in SAR images. (b) Examples of rotations of ship in SAR images.

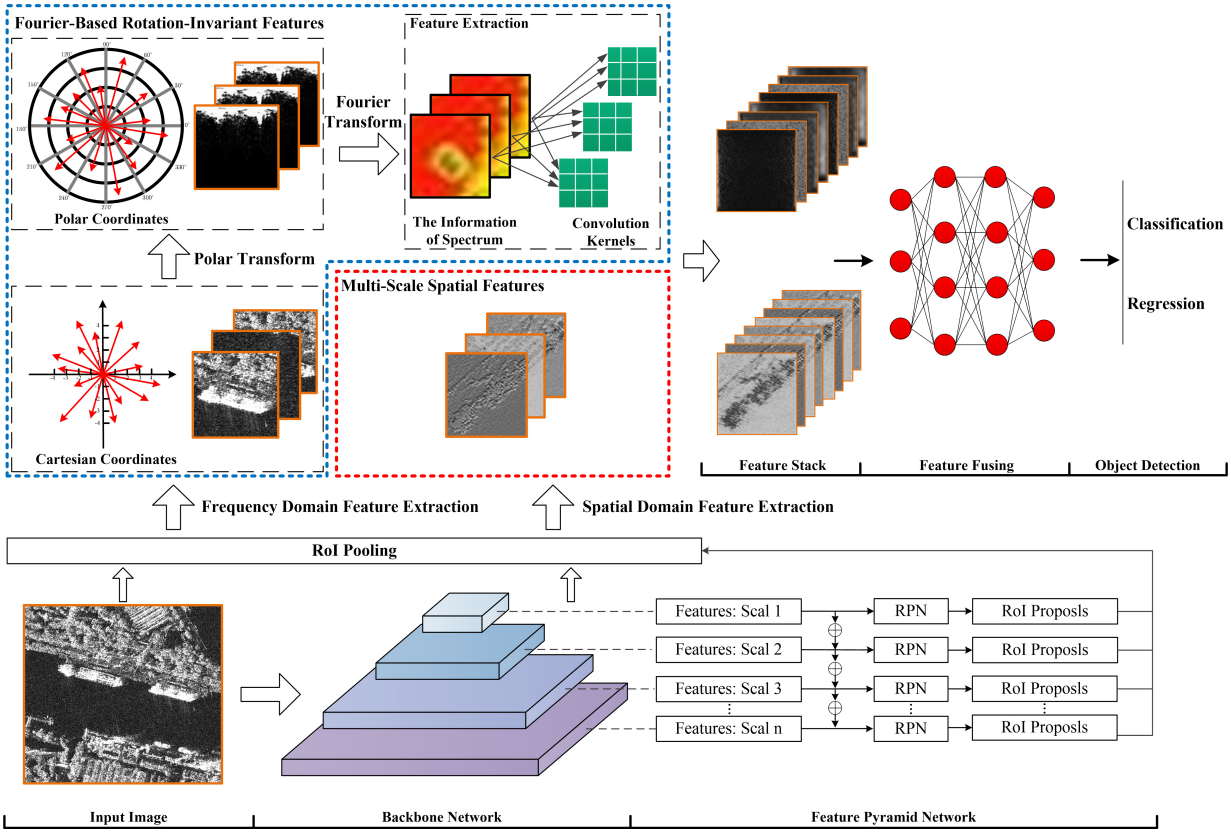

Fig. 2. Overview of the processing scheme. It mainly includes three parts: 1) multiscale feature extraction based on FPN; 2) Fourier-based rotation-invariant features extraction; and 3) feature fusion and object detection.

complementary properties of multidimensional domains. The each step will be described fully next.

B. Multiscale Spatial Features Extraction

In the convolutional neural network, by using the characteristics of convolution operation, it can independently extract the features of the marked data, and has the advantages of being fast and efficient, which can effectively avoid the design of traditional complex features and greatly improve the detection accuracy. Different from BP neural network, CNN adopts a parameter sharing mechanism, which can realize the training of deeper network, so as to achieve the purpose of deep learning. In this work, we take advantage of CNN to obtain the spatial features from SAR images through the backbone networks. He *et al.* [32] proposed a deep residual network for image recognition to ease the training process. Therefore, we adopt an extractor based on ResNet101 to extract the spatial features of SAR images. The residual structure in ResNet101 is called bottleneck, shown in Fig. 3.

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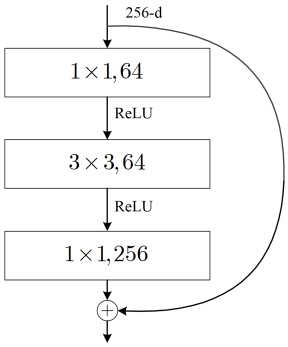

Fig. 3. Architecture of residual bottleneck.

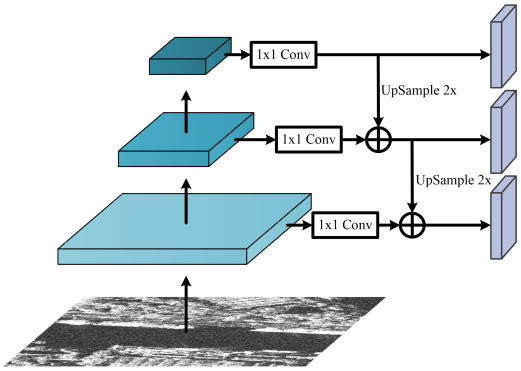

Fig. 4. Architecture of feature pyramid network.

The structure of ResNet101 extractor is divided into five convolution stages. Each stage outputs a feature of different scales, represented by {conv1, conv2, conv3, conv4, conv5}. Each stage consists of several conventional convolution layers or bottleneck.

Moreover, the low-level features lack semantic information, but the position information is accurate, which means it is suitable for small-scale ships detection. On the contrary, the high-level features contain rich semantic information, but the location information is poor, which indicates it fits for detecting large-scale ships. Besides, the features of different layers could be imbalanced. Therefore, the FPN utilizes the information of each layer in the CNN network to generate the finally combined expression features, shown in Fig. 4. The architecture of feature pyramid network can be divided into down-top pathway, top-down pathway, and lateral connection. In the down-top pathway, the multilayer features are extracted by the backbone network as mentioned above. The features pyramid designs a top-down pathway which refers to the high-level information of the previous layer by up-sampling as the input when processing each layer of information. The up-sampled map is merged with the feature map generated from the bottom up which undergoes the 1×1 Conv operate to reduce channel dimensions. Finally, the merged feature maps are appended to generate a fine-grained features pyramid.

C. Fourier-Based Rotation-Invariant Features Extraction in Polar Coordinate

1) Polar Fourier Transform: In the above discussions, only CNN is used to extract the spatial features of ship targets, and the feature information of other dimensions of SAR ship targets is ignored. In practical applications, ship targets often suffer from various complex deformations as well. In this article, we use polar coordinate transformation to transform the rotation transformation in Cartesian coordinates into translation, and transform the translation invariance of the Fourier transform to accommodate SAR ship targets arbitrary rotation angles.

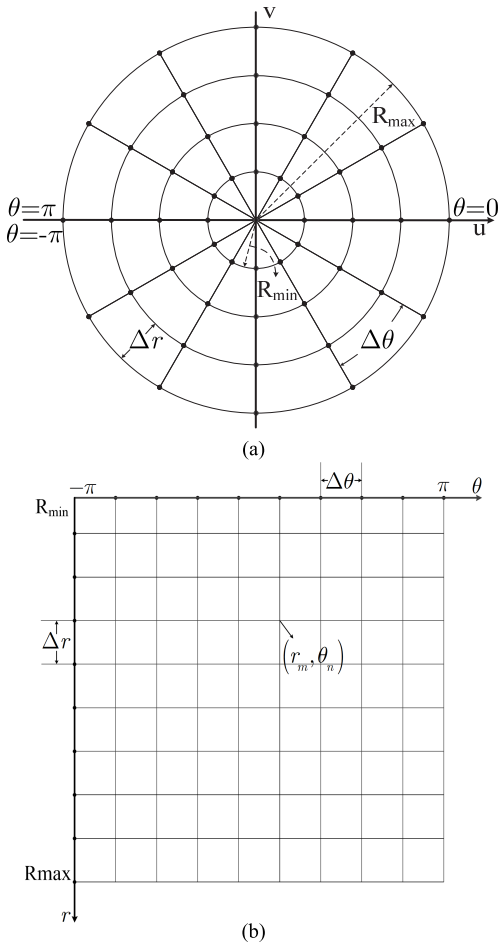

Fig. 5. Sketch of a transformation from Cartesian coordinates to polar coordinates. (a) Polar grids. (b) Geometry of the polar image.

The position of a pixel in a 2-D image can be represented by Cartesian coordinates or by polar coordinates, and the transformation relationship is shown in Fig. 5. The torus and spoke lines in the Cartesian coordinate of the Fig. 5(a) are mapped to the transverse lines and longitudinal lines in the polar coordinate of the Fig. 5(b), respectively.

Let the coordinates in Cartesian coordinate be represented as (x, y), and the polar coordinate transformation is defined as

$$\begin{cases} r = \sqrt{x^2 + y^2} \\ \theta = \tan^{-1}(y/x) \end{cases} \tag{1}$$

where r represents the distance from the point to the origin in Cartesian coordinates and θ is the angle between x axis and the line that the point is connected to the origin which takes the counterclockwise as positive.

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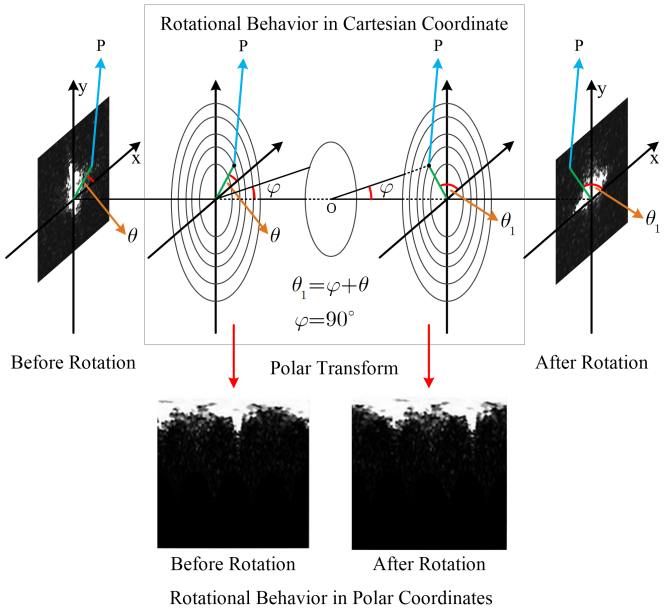

Fig. 6. Rotation behavior in Cartesian coordinates and polar coordinates.

It is noted that in our article, we first perform polar Fourier transform. In doing so, the rotation can be transformed as the translation by using polar Fourier transformation. In order to clearly describe the problem of the rotation of ship targets and clarify the solution, we first denote the SAR image as $\mathbf{I_1}(x, y)$ where (x, y) represents the location of a given pixel. Let $\mathbf{I}_2$ be a rotated replica of $\mathbf{I}_1$, given by

$$\mathbf{I}_2(x, y) = \mathbf{I}_1(x\cos\varphi - y\sin\varphi, x\sin\varphi + y\cos\varphi) \tag{2}$$

where φ is the rotation angle. After the transformation by polar coordinates, they are represented by $\mathbf{p_1}(r, \theta)$ and $\mathbf{p_2}(r, \theta)$, and the relation is

$$\mathbf{p_2}(r, \theta) = \mathbf{p}_1(r, \theta + \varphi). \tag{3}$$

From (3), the rotation in Cartesian coordinates can be transformed into φ translation in polar coordinates, and Fig. 6 illustrates the nature of polar coordinate transformations.

Therefore, as long as the translation of the image is eliminated, the rotation-invariant feature maps can be obtained. The rotation invariance can be achieved by taking the Fourier transform in polar coordinates. Let $F(\cdot)$ indicates the Fourier transform operation, $\mathbf{F_1}(u, \upsilon)$ and $\mathbf{F_2}(u, \upsilon)$ are the Fourier transforms of $\mathbf{p_1}(r, \theta)$ and $\mathbf{p_2}(r, \theta)$, respectively where u and υ are frequency-domain variables. From the shift properties of Fourier transform, the translation behavior in polar coordinates is defined as

$$\begin{aligned}\mathbf{F_2}(u, \upsilon) &= F(\mathbf{p_2}(r, \theta)) = F(\mathbf{p_1}(r, \theta + \varphi)) \\ &= \mathbf{F_1}(u, \upsilon) \cdot \exp(-j2\pi\varphi\upsilon).\end{aligned} \tag{4}$$

Suppose $\mathbf{M_1}(u, \upsilon)$ and $\mathbf{M_2}(u, \upsilon)$ are the amplitude spectrums of $\mathbf{F_1}(u, \upsilon)$ and $\mathbf{F_2}(u, \upsilon)$, and they are related by

$$\mathbf{M_2}(u, \upsilon) = \|\mathbf{F_1}(u, \upsilon) \cdot \exp(-j2\pi\varphi\upsilon)\| = \mathbf{M_1}(u, \upsilon). \tag{5}$$

Equation (5) shows that the amplitude spectrum is independent of the translation. Therefore, the rotation behavior of the image can be transformed into translation in the direction of θ by polar transform, and then the rotational invariance by the Fourier transform is realized. This transformation process is called polar Fourier transform.

However, in SAR ship detection, due to the lack of time-frequency locating ability, the frequency-domain characteristic information corresponding to the SAR ship targets cannot be directly extracted accurately. Therefore, in this work,-we first exploit the region proposal network (RPN) to generate a series of candidate regions with location information and category information. And then the location information of candidate region can be mapped into the feature maps and original image to obtain the corresponding ship targets image. Therefore, the frequency-domain feature information can be obtained by performing the Fourier transform, which realizes the time-frequency features alignment for the SAR ship target. The implementation process is shown in Fig. 2. Based on the above analysis, we obtained the rotation-invariant characteristics of the ship targets through polar Fourier transform. We then can extract the frequency-domain rotation-invariant feature via CNN.

2) Fourier-Based Rotation-Invariant Features Extraction: First, some notations are defined to clarify the solution in this section. The input SAR image is denoted by $\mathbf{I}(x, y)$ where (x, y) refers to the location of a given pixel. According to the Section B, the spatial features have been obtained, denoted by $\mathbf{X_S}$. After extracting shared feature maps with a CNN, the RPN generates a set of rectangular candidate areas containing location information and category score. The obtained proposals are mapped to the feature maps to obtain the spatial features $\mathbf{X_{S_sub}}$. At the same time, the proposals are also mapped to the original SAR image to obtain the corresponding image information $\mathbf{I_{sub}}(x, y)$. Through the polar Fourier transform, $\mathbf{I_{sub}}(x, y)$ is conducted to obtain its rotation invariant information $\mathbf{M_{sub}}(u, \upsilon)$. Therefore, we can design a CNN network to extract its rotation-invariant features of the proposals.

With these definitions, the output in the $l-$th layer of the CNN is

$$\mathbf{z}^{(l)}_{\mathbf{F_sub}} = \begin{cases} h_{\mathbf{w}^l\mathbf{b}^l}(\mathbf{M_{sub}}(u, \upsilon)), & l = 1 \\ h_{\mathbf{w}^l\mathbf{b}^l}\left(\mathbf{z}^{(l-1)}_{\mathbf{F_sub}}\right), & l = 2, \ldots, n \end{cases} \tag{6}$$

where $h(\cdot)$ represents the convolutional operation, $\{\mathbf{w}^l\}_{l=1}^n$ and $\{\mathbf{b}^l\}_{l=1}^n$ are defined as the weights and the biases of all layers $(l = 1, \ldots, n)$. In order to speed up the convergence of network, batch normalization (BN) operation [33] is carried out after each convolutional layer in the network, given by

$$\mathbf{z}^{(l)}_{\mathbf{BNF_sub}} = \gamma \frac{\mathbf{z}^{(l)}_{\mathbf{F_sub}} - m\left(\mathbf{z}^{(l)}_{\mathbf{F_sub}}\right)}{\sqrt{\sigma^2\left(\mathbf{z}^{(l)}_{\mathbf{F_sub}}\right) + \varepsilon}} + \beta \tag{7}$$

where $\mathbf{z}^{(l)}_{\mathbf{F_sub}}$ is the input vector, $m(\mathbf{z}^{(l)}_{\mathbf{F_sub}})$ is the mean value of $\mathbf{z}^{(l)}_{\mathbf{F_sub}}$, $\sigma^2(\mathbf{z}^{(l)}_{\mathbf{F_sub}})$ is the variance value of $\mathbf{z}^{(l)}_{\mathbf{F_sub}}$, γ and β are the hyper parameters, and ε is a small constant to ensure that the denominator is not zero. After that, $\mathbf{z}^{(l)}_{\mathbf{BNF_sub}}$ is

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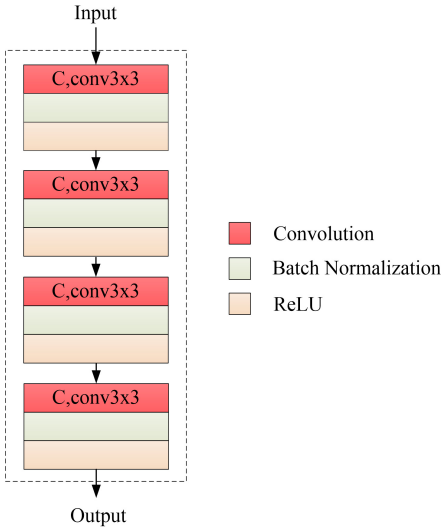

Fig. 7. Structure of feature extraction network for frequency domain.

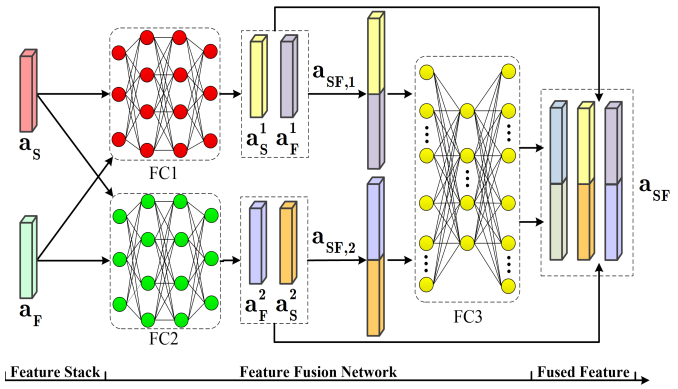

Fig. 8. Network structure of multidimensional domain features fusion. $\mathbf{a}_S^1$ and $\mathbf{a}_F^1$ are the feature vectors of $\mathbf{a_S}$ and $\mathbf{a_F}$ that derived by FC1; $\mathbf{a}_F^2$ and $\mathbf{a}_S^2$ are the feature vectors of $\mathbf{a_F}$ and $\mathbf{a_S}$ that derived by FC2.

non-linearized by activation function as

$$\mathbf{X}_{\mathbf{F_sub}}^{(l)} = f\left(\mathbf{z}_{\mathbf{BNF_sub}}^{(l)}\right) \tag{8}$$

where $f(\cdot)$ adopts ReLU as the activation function, which is defined by

$$f(x) = \begin{cases} 0, & x \le 0 \\ x, & x > 0. \end{cases} \tag{9}$$

The structure of feature extraction network for the frequency domain is shown as Fig. 7, which consists of four layers with the sizes of $(16, 32, 64, 64)$. Each of these layers is designed according to the above process and the convolution kernel with the size of 3×3. By using the polar Fourier transform and feature extraction, the rotation-invariant features $\mathbf{X_{F_sub}}$ of the proposals are obtained in the frequency domain.

D. Feature Fusion Network

The spatial features of different scales and the rotation-invariant features in the frequency domain are now obtained. The feature fusion based on splicing has been widely used. However, this method cannot comprehensively express the characteristics of each dimension domain, and its ability to fuse the features of multidimension domain is limited.

In order to exploit the features better, we design a more refined feature fusion network that can comprehensively represent the features of each dimension domain. In [34], a feature fusion module is proposed called cross fusion that can obtain more compact feature representations in different modalities through updating the parameters of different networks interactively. The cross fusion achieves a more full feature fusion. On this basis, we propose a network suitable for multidimensional domain features fusion, shown in Fig. 8.

In order to ensure that the dimensions of spatial and frequency features are consistent, we flatten the spatial features $\mathbf{X_{S_sub}}$ into a 1-D vector and the resulting output features size is 1024, where FC represents the fully connected layer. Similarly, we flatten the frequency-domain features $\mathbf{X_{F_sub}}$ to obtain the features vectors with the same dimension. The representations are

$$\mathbf{a_S} = \mathrm{FC}\left(\mathrm{Flatten}\left(\mathbf{X_{S_sub}}\right)\right) \tag{10}$$

$$\mathbf{a_F} = \mathrm{Flatten}\left(\mathbf{X_{F_sub}}\right) \tag{11}$$

where a_S and a_F respectively represent the output feature vector of spatial and frequency domains. After that, the fusion operation is conducted, given by

$$\begin{aligned} \mathbf{a_{SF,1}} &= [\mathrm{FC}_1(\mathbf{a_S}), \quad \mathrm{FC}_1(\mathbf{a_F})] \\ \mathbf{a_{SF,2}} &= [\mathrm{FC}_2(\mathbf{a_F}), \quad \mathrm{FC}_2(\mathbf{a_S})] \\ \mathbf{a_{SF}} &= \mathrm{Flatten}\left(\begin{bmatrix} \mathrm{FC}_3(\mathbf{a_{SF,1}}), & \mathrm{FC}_3(\mathbf{a_{SF,2}}) \\ \mathrm{FC}_1(\mathbf{a_S}), & \mathrm{FC}_2(\mathbf{a_S}) \\ \mathrm{FC}_1(\mathbf{a_F}), & \mathrm{FC}_2(\mathbf{a_F}) \end{bmatrix}\right) \end{aligned} \tag{12}$$

where $[\cdot, \cdot]$ denotes to splice the two feature vectors together and each row in a_{SF} shares the weight. The cross between network and features makes the output features more compact while preserving the characteristic of each dimension domain. By using this cross, a better and more effective expression of spatial frequency feature is realized.

E. Loss Function

Similar to other classical two-stage target detection networks, both RPN and detection networks are optimized by using multitask losses. The loss function is defined by

$$L\{p_i, t_i\} = \frac{1}{N_{\mathrm{cls}}} \sum_i L_{\mathrm{cls}}\left(p_i, p_i^*\right) + \lambda \frac{1}{N_{\mathrm{reg}}} \sum_i p_i^* L_{\mathrm{reg}}\left(t_i, t_i^*\right) \tag{13}$$

where λ is a weighting parameter, p_i is the prediction probability of the $i-$th anchor, p_i^* is the label of ground-truth and it is 1 when the anchor is positive, or else p_i^* is 0, $t_i = \{t_x, t_y, t_w, t_h\}$ is the parameterized coordinate vectors where t_x, t_y represent the coordinates of the center point of the candidate box and t_w, t_h represent the width and height of the candidate box, and t_i^* is the coordinate vector of the corresponding ground-truth, N_{cls} and N_{reg} are the numbers of mini-batch in the training stage, L_{cls} and L_{reg} are defined as

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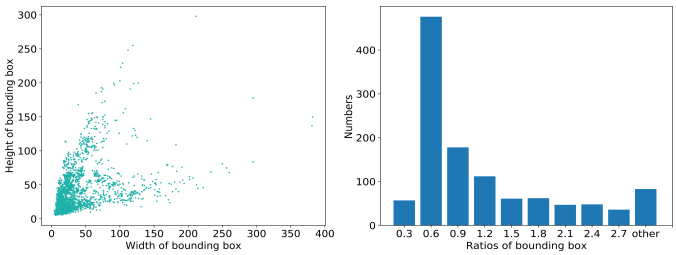
TABLE I
INFORMATION ABOUT THE SSDD DATA SET

Sensors	Polarization	Resolution	Position
Sentinel-1 RadarSat-2 TerraSAR-X	HH, VV VH, HV	1m-15m	inshore offshore



(a) (b)

Fig. 9. Distribution of SSDD. (a) Bounding box sizes distribution of in SSDD. (b) Bounding box aspect ratio distribution of in SSDD.

classification and regression losses, respectively. Cross entropy is utilized as the classification loss, which is given by

$$L_{\text{cls}}\left(p_i, p_i^*\right) = -\log\left(p_i^* p_i + (1 - p_i^*)(1 - p_i)\right). \quad (14)$$

The regression loss function is defined by

$$L_{\text{reg}}\left(t_i, t_i^*\right) = R\left(t_i - t_i^*\right) \quad (15)$$

where R is smooth L1 function as

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2, & |x| \le 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \quad (16)$$

III. EXPERIMENTAL RESULTS AND EVALUATION

In this section, the performance of our method is evaluated by using the open-source data set, SSDD. First, we introduce the SSDD data set and the setting of our experiments, and then, the evaluation metrics are described. The next part focuses on the explorations of our method including the influence of backbone network, multiscale features, and rotation-invariant features. Finally, we compare the proposed method with other CNN-based methods to verify the superiority of the proposed method.

A. Data Set Description and Settings

The establishment of a large and representative database is the basis of deep learning-based object detection. The SSDD data were mainly derived from Radarsat-2, TerraSAR-X, and Sentinel-1 sensors with four polarization modes including HH, HV, VV, and VH. The resolution of SAR images ranges from 1m to 15m, and the details of data information are summarized in Table I.

The SSDD consists of 1160 SAR images and 2456 ship targets, and each image contains an average of 2.12 ship targets. The distribution of SSDD is shown in Fig. 9, where Fig. 9(a) and (b) represents the height-width and height-width ratio distribution of each ship target, respectively. In addition, the SAR images of serial number ending with 0 and 9 in SSDD are unified as the test set for result analysis to standardize the research benchmark of SSDD.

In this article, the pre-trained ResNet101 on the ImageNet [35] is adopted as the backbone network and the learning rate is set as 0.001. Besides, the max iteration is 20 000 and we save checkpoints every 5000 steps. The experiments are executed in TensorFlow [36] on the computer with Intel(R) I7-9700 CPU, NVIDIA GTX1660Ti GPU. The operating system is Ubuntu16.04 with Cuda10 and Cudnn 8.

B. Evaluation Criteria

In order to evaluate the detection performance of the proposed method, several indicators are utilized for evaluations such as precision rate, recall rate, F1-score, and mean average precision (mAP).

The precision is the ratio of the ships detected correctly in all detected ships and it is defined as

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \quad (17)$$

where TP is true positives which represents the numbers of real ships are detected and FP is false positive representing false alarm.

The recall rate is a measure of coverage, which represents the ratio of the ships being detected in all ground truths, which is given by

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \quad (18)$$

where FN is false negatives that can be understood as missed detection.

The F1-score evaluates the precision rate and recall rate, and it is

$$\text{F1-score} = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (19)$$

The mAP is defined by

$$\text{mAP} = \int_0^1 P(R) dR \quad (20)$$

where P(R) is the curve of Precision-Recall.

C. Evaluations of the Proposed Method

The section evaluates the effectiveness of the proposed method, and Faster RCNN is used as a baseline since the proposed method has a similar structure to Faster RCNN. In addition, experiments are carried out in both offshore and inshore scenarios to analyze the detection performances of the proposed method in different scenarios.

1) Effect of Backbone Network: Because different backbone networks have different data processing results, this section analyzes and compares different backbone network models on Faster RCNN to evaluate the influences of backbone networks. All the backbone networks are pre-trained on ImageNet, and Table II shows the detection results of Faster RCNN. From the table, the backbone network of ResNet101 has the best detection performance and the mAP, the F1-score, the precision rate, and the recall rate are 90.8%, 91.0%, 93.4%,

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TABLE II
DETECTION RESULTS ON DIFFERENT BACKBONE

Backbone	mAP	F1-score	P	R
VGG16	0.888	0.902	0.933	0.872
ResNet50	0.894	0.904	0.917	0.892
ResNet101	0.908	0.910	0.934	0.888

TABLE III
DETECTION RESULTS OF METHODS ON OFFSHORE SHIPS

Method	FPN	RIF	mAP	F1-score	P	R
Faster RCNN			0.985	0.981	0.976	0.987
Faster RCNN with FPN	✓		0.989	0.978	0.976	0.981
Faster RCNN with RIF		✓	0.987	0.985	0.984	0.987
Proposed method with FPN and RIF	✓	✓	0.994	0.988	0.992	0.984

TABLE IV
DETECTION RESULTS OF METHODS ON INSHORE SHIPS

Method	FPN	RIF	mAP	F1-score	P	R
Faster RCNN			0.719	0.758	0.823	0.703
Faster RCNN with FPN	✓		0.811	0.775	0.870	0.698
Faster RCNN with RIF		✓	0.745	0.791	0.885	0.715
Proposed method with FPN and RIF	✓	✓	0.836	0.794	0.843	0.750

and 88.8%, respectively. This might be due to the fact that Res101 has a deep network structure and is easier to converge than VGG16. Therefore, in order to obtain better detection performance and illustrate the effectiveness of the proposed method, the following experimental analysis is all based on the backbone network of Res101.

2) Effect of Frequency-Domain Features: We conduct a comparison of different methods in offshore and inshore scenes, where FF indicates polar coordinate transformation is not performed on ship targets, and only frequency-domain features are extracted. Tables III and IV respectively show the detection results of different methods.

In Table III, the detection performance of the four methods is roughly similar and there are small differences in each detection indicator since the background of offshore ships is relatively simple with less interference. Even so, the comprehensive utilization of spatial frequency features slightly improves the performance compared with that of single-dimensional domain. The method that only takes advantage of FPN or FF is similar to the Faster RCNN, whereas the precision rate, the F1-score, and mAP are improved by 1.6%, 0.7%, and 0.9%, respectively, by simultaneously using FPN and FF.

Different from offshore ships, the background clutter of inshore ships is more complex. Additionally, the quays and buildings on the shore induce great disturbances to the detection of the inshore ships and the performance of SAR ship detection methods generally degrades. In Table IV, the indicators of different methods have dropped significantly. However, the detection performance is improved obviously through extracting the complementary features in the frequency domain. By introducing the FF module, the mAP, the F1-score, the precision rate, and the recall rate are 74.5%, 79.1%, 88.5%, and 71.5% with the improvements of 2.6%, 3.3%, 6.2%, and 1.2% compared with Faster RCNN, respectively. This result shows that utilizing the complementary features of the SAR ship in the frequency domain complements the spatial features, which is helpful to improve the SAR ship detection performance under the complex background.

The PR curves of different methods are shown in Fig. 10, where Fig. 10(a) and (b) represents the offshore and inshore test results, respectively. In Fig 10(a), the PR curves almost coincide with each other and there are only small differences. In Fig. 10(b), the performance difference is more obvious on inshore ships. The blue curve is the PR curve of Faster RCNN and it is shown that the detection performance is the worst. With the utilization of FPN and FF, the detection performance is the best because we make a full use of the spatial multiscale feature and the frequency-domain feature.

3) Effect of Rotation-Invariant Features: The simple feature extraction of frequency-domain information cannot fully explore the potential features of SAR ships. To improve on that, the frequency-domain rotation-invariant feature (RIF) is further extracted by using the polar Fourier transform in the frequency domain, where RIF indicates the frequency-domain rotation-invariant feature is used. The detection results are shown in Tables V and VI. Similar to Table III, there is a small difference among different algorithms in Table V. Compared with Table III, the introduction of RIF does not significantly help the detection results. This is because under the simple background, the detection performance of each method has been excellent, which may reach the bottleneck and limit the improvement of detection performance.

In the inshore ships detection, even under the influence of the surrounding coastal environment, the proposed method with FPN and RIF still maintains a high detection performance and the indicators are 85.2%, 82.7%, 90.3%, and 76.2%, respectively, as shown in Table VI. Compared with Faster

TABLE V
DETECTION RESULTS OF METHODS ON OFFSHORE SHIPS

Method	FPN	RIF	mAP	F1-score	P	R
Faster RCNN			0.985	0.981	0.976	0.987
Faster RCNN with FPN	✓		0.989	0.978	0.976	0.981
Faster RCNN with RIF		✓	0.985	0.984	0.979	0.989
Proposed method with FPN and RIF	✓	✓	0.996	0.992	0.992	0.992

TABLE VI
DETECTION RESULTS OF METHODS ON INSHORE SHIPS

Method	FPN	RIF	mAP	F1-score	P	R
Faster RCNN			0.719	0.758	0.823	0.703
Faster RCNN with FPN	✓		0.811	0.775	0.870	0.698
Faster RCNN with RIF		✓	0.761	0.787	0.902	0.698
Proposed method with FPN and RIF	✓	✓	0.852	0.827	0.903	0.762

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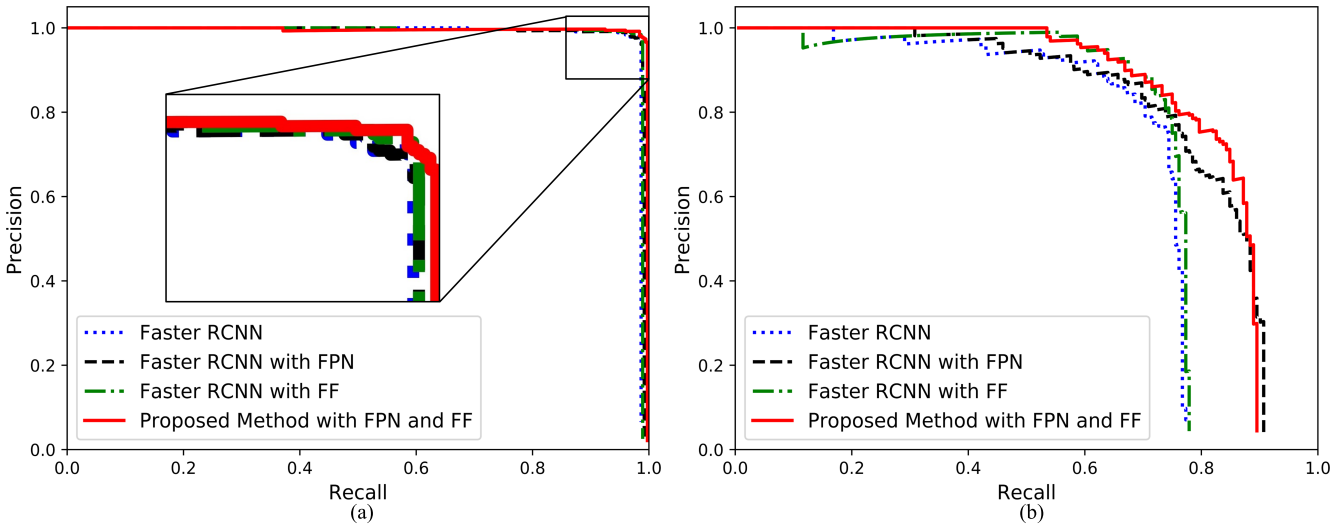

Fig. 10. PR curves of different methods on offshore and inshore ships. (a) PR curves of different methods on offshore. (b) PR curves of different methods on inshore.

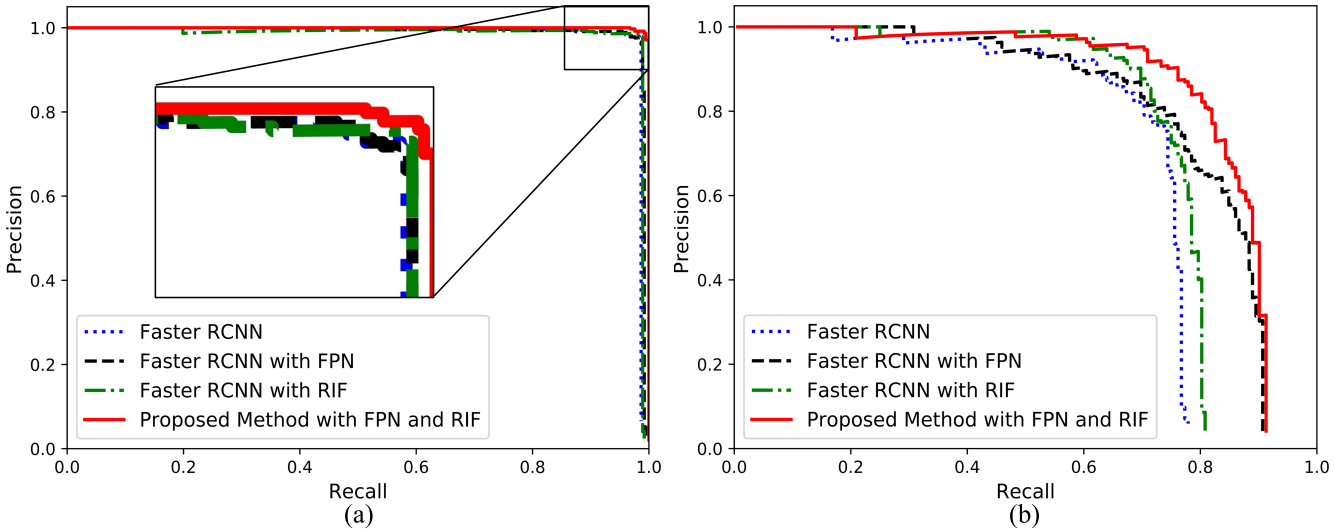

Fig. 11. PR curves of different methods on offshore and inshore ships. (a) PR curves of different methods on offshore. (b) PR curves of different methods on inshore.

RCNN, the mAP, the F1-score, the precision rate, and the recall rate are improved by 13.3%, 6.9%, 8%, and 5.9%, respectively. This is because we make a full use of the rotation-invariant features of ship targets in the frequency domain and utilize appropriate feature fusion and strategies. Tables IV and VI show that the detection performance of RIF is also improved compared with using the FF. Among them, the precision rate is the most obvious improvement. It suggests that the false alarm can be effectively reduced by applying RIF.

Fig. 11 shows the PR curves of different methods and Fig. 11(a) and (b) shows the detection results in offshore and inshore ships, respectively. The PR curves of these methods are very close in Fig. 11(a) suggesting a satisfactory performance in offshore ship detection. When the background becomes complex, these methods have obvious differences, which reflects the effectiveness of the proposed method.

From the above analysis, it is concluded that false alarm can be effectively reduced by using frequency-domain feature. The comprehensive utilization of rotation invariant feature and multiscale feature can effectively improve the SAR ship detection performance. In the areas with dense small ship targets, false alarms can be effectively reduced by introducing frequency-domain features.

4) Detection Results of Different Methods: The detection results using SSDD based on Faster RCNN and other methods are shown in Figs. 12 and 13. Fig. 12 shows the detection results of different methods in the offshore scene. In Fig. 12(a), the pink rectangles are the ground truths of SAR ship images. Fig. 12(b) shows the detection results of Faster RCNN in which the detected ship targets are marked with red rectangles. Similarly, Fig. 12(c) shows the detection results of Faster RCNN that uses FPN. Fig. 12(d) depicts the detection results

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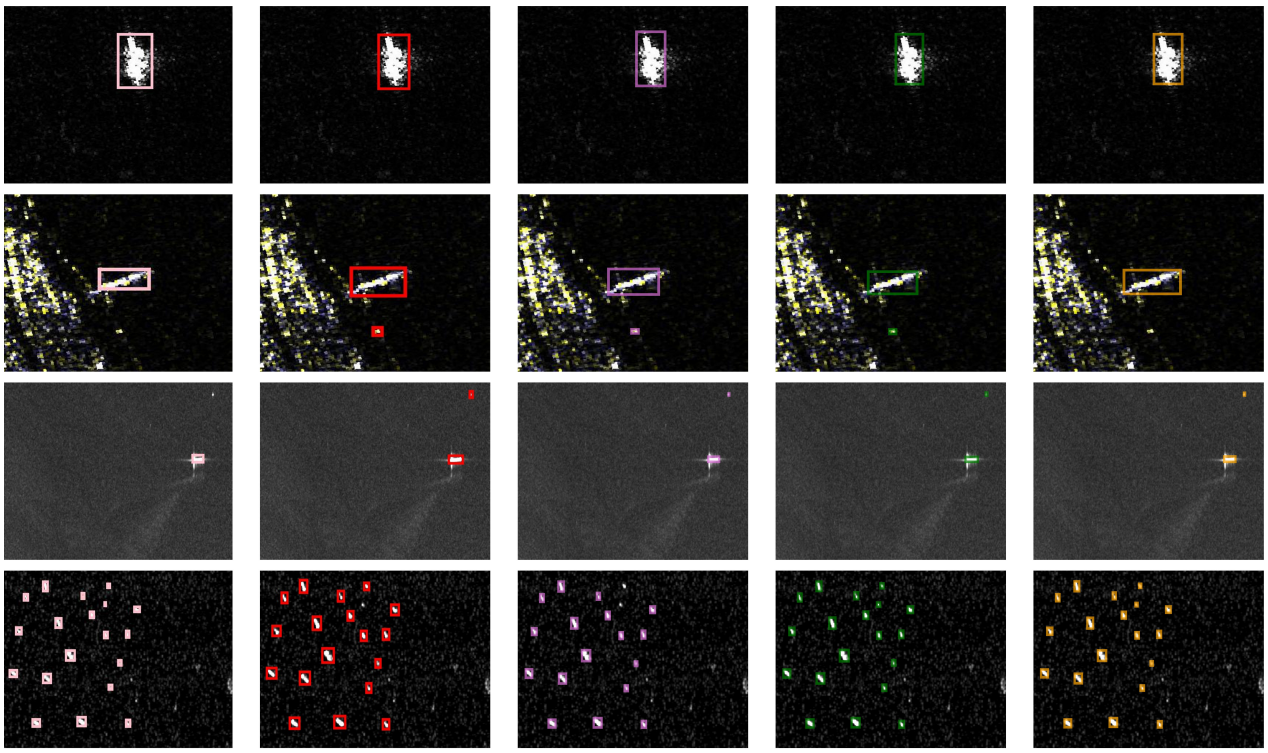

Fig. 12. Detection results of different methods in offshore scene. (a) Ground truth. (b) Faster RCNN. (c) Faster RCNN with FPN. (d) Faster RCNN with FPN and FF. (e) Proposed method.

of Faster RCNN with FPN and FF. The detection results of the proposed method are shown in Fig. 12(e).

It is seen from Fig. 12 that both the compact small targets and single targets can be detected by the proposed method. Compared with other methods, the proposed method can reduce false alarm to a certain extent in the offshore scene.

Fig. 13 shows the detection results of different methods in the inshore scene. Similar to Fig. 12, Fig. 13(a) is the ground truth and Fig. 13(b) and (c) shows the detection results of different methods. The problems of false alarms and missed detections are more obvious in the inshore scene. It is easy to generate false alarms due to the great similarity between the inshore buildings and the ship targets. In Fig. 13(c), there are some false alarms and missed detections when the Faster RCNN only uses FPN. With the addition of frequency-domain features, the false alarms are reduced, shown in Fig. 13(d). It can be seen from Fig. 13(e) that, even if SAR ship targets have a high degree of similarity with background clutter such as inshore buildings, a relatively accurate detection result is obtained, compared with the ground truth. In both offshore and inshore scenes, the proposed method presents a superior detection performance for SAR ships.

D. Comparison With Other Methods

In this section, we further conduct the comparison of computational times, shown in Fig. 14. Compared with existing methods, it is obvious that the mAP of the proposed method is significantly improved. According to Fig. 14, although the detection speed decreases with introducing FPN in Faster RCNN, its mAP improves by nearly 4%. On this basis, with simultaneously exploiting spatial-frequency complementary features of ship targets, our proposed method obtains better detection performances at the expense of the detection speed loss less than 0.1s.

TABLE VII
RESULTS OF DIFFERENT METHODS FOR SAR SHIP DETECTION

Method	mAP	F1-score	P	R
Faster RCNN	0.908	0.910	0.934	0.888
Faster RCNN with FPN	0.945	0.915	0.956	0.877
SSD	0.918	0.875	0.879	0.872
YOLO v3	0.919	0.893	0.914	0.872
Proposed Method	0.962	0.939	0.946	0.932

We now compare CNN-based object detection methods including two-stage algorithm and one-stage algorithm with the proposed method in the whole test set and Table VII shows the comparison results. It is seen from the table that the two-stage object detection algorithms have a better precision rate while maintaining the recall rate. The results show that the mAP of the proposed method reaches 96.2%, which is 5.4% higher than the Faster RCNN and nearly 4.5% higher than the SSD and YOLO v3. This indicates that the multidimensional domain fusion feature map with abundant spatial multiscale information and frequency-domain rotation invariant information is effective for SAR ship detection. This is because we make a full use of the complementary features of

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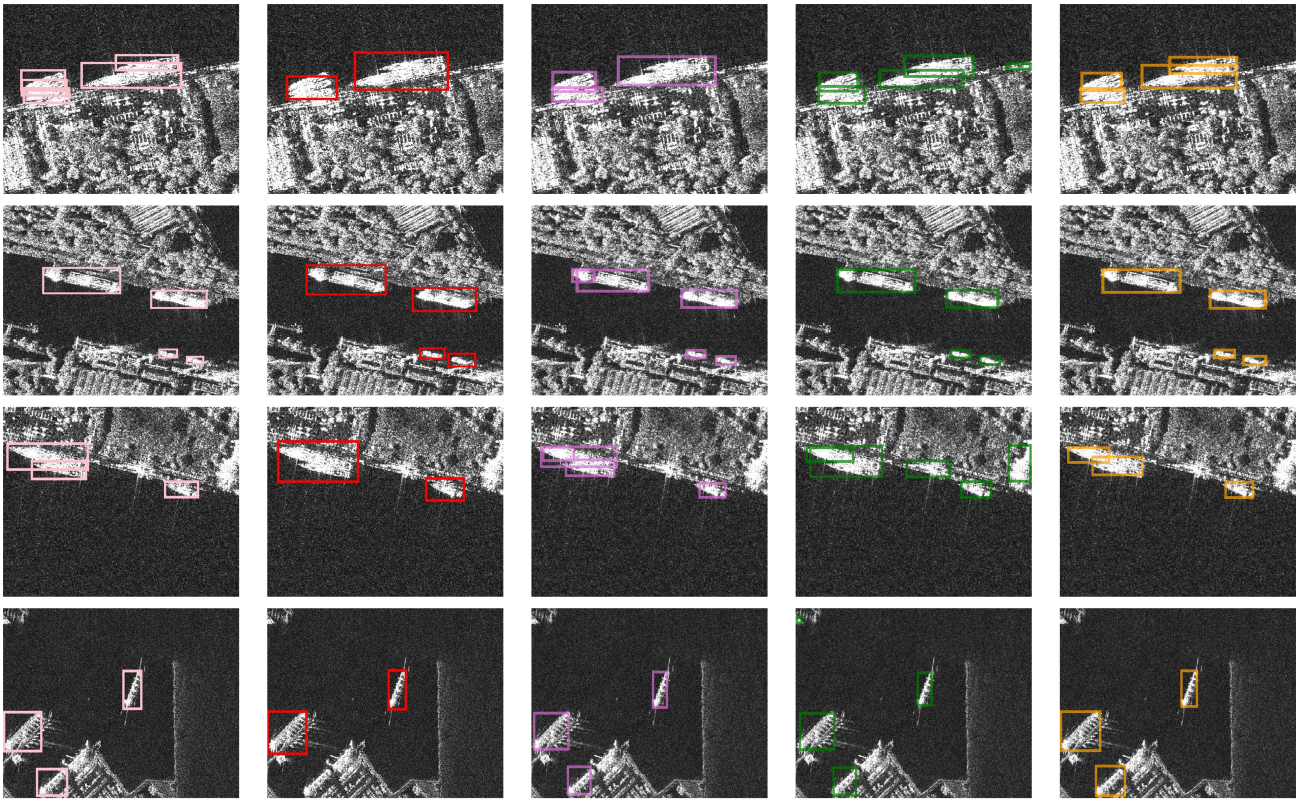

Fig. 13. Detection results of different methods in inshore scene. (a) Ground truth. (b) Faster RCNN. (c) Faster RCNN with FPN. (d) Faster RCNN with FPN and FF. (e) Proposed method.

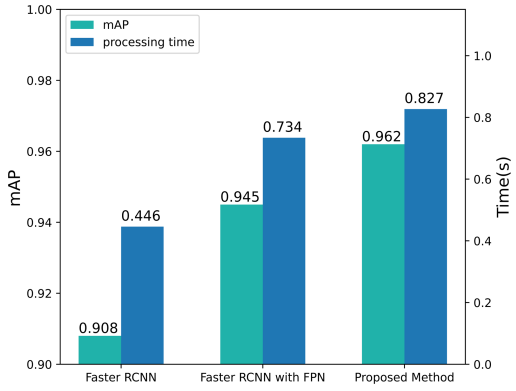

Fig. 14. Bar graph of mAP (accuracy) and processing time (speed) of different methods.

spatial frequency for ship targets and the appropriate strategy of feature fusion is also conducted.

IV. Conclusion

In this article, a multidimensional domain deep learning network for SAR ship detection is proposed. In the developments of the proposed approach, based on Faster RCNN, the multidimensional domain information of SAR ship image is deeply explored and the complementary features of spatial and frequency are used to improve the detection performance, where FPN is utilized to obtain multiscale features in the spatial domain and the polar Fourier transform is used to produce frequency features. With the experiments using the SSDD data set and comparisons with the existing methods, the proposed method demonstrates a superior performance in both offshore and inshore scenes.

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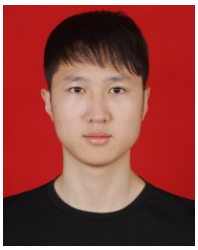

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